

Sprint 48: GLM Standardized Evaluation - Abdollah Ghazvanchahi

GLM Standardized Evaluation and Fine-Tuning Pipeline Fit Assessment

Executive Summary

Sprint 48 conducted a standardized evaluation of “GLM-4-9B” against the current baseline, “Qwen/Qwen3-8B”, for EN↔CMN and EN↔YUE translation using the FLORES-200 devtest split. The purpose was to determine whether GLM should be adopted alongside the baseline, replace it, or be ruled out before any fine-tuning work begins.

The evaluation shows a mixed but informative outcome. GLM-4-9B outperforms Qwen3-8B on the Mandarin directions (EN→CMN and CMN→EN) and achieves a small advantage on YUE→EN by COMET. However, Qwen3-8B remains clearly stronger on EN→YUE, the most important direction for direct Cantonese generation, with better COMET, BLEU, and chrF.

Given these results, GLM-4-9B should not replace Qwen3-8B as the primary model at this stage. However, it performs strongly enough to remain a serious candidate for the pipeline, especially for Mandarin directions and as a fallback or complementary model.

Final recommendation:

Adopt GLM-4-9B alongside Qwen3-8B, but do not replace Qwen3-8B as the primary baseline before blind human review and pipeline-fit checks are completed.

1. Fixed Baseline Configuration

To satisfy the sprint requirement for a single reproducible benchmark, Sprint 48 used one locked baseline configuration for all comparisons.

1.1 Baseline model

- Model family: Qwen
- Model variant: Qwen3-8B
- Role: current standardized baseline for Sprint 48 comparison against GLM

1.2 Baseline quantization and loading

- Quantization method: 4-bit bitsandbytes
- Runtime/loading framework: Unsloth `FastLanguageModel`

The baseline model was initialized as follows:

```
model, tokenizer = FastLanguageModel.from_pretrained(
    model_name = MODEL_NAME,
    max_seq_length = 4096,
    dtype = None,
    load_in_4bit = True,
)
FastLanguageModel.for_inference(model)
```

1.2.1 Role of Unsloth in the baseline

In this project, Unsloth was used as the model-loading and inference optimization layer for the baseline setup. Its role was to:

- load the model in 4-bit quantized form
- reduce memory usage so the model could run within available Colab/Kaggle/GPU constraints
- speed up loading, serving, and inference
- provide a workflow that is also compatible with later fine-tuning experiments, including LoRA/QLoRA-style adaptation

For the purposes of this evaluation, Unsloth should be described as:

- an optimized inference and fine-tuning wrapper
- used to make large language models more practical under limited hardware
- responsible for handling the project's chosen 4-bit loading path
- part of the reproducible environment in which baseline and GLM comparisons were run

1.3 Baseline prompt template

The exact fixed prompt template used for baseline generation was:

```

prompt = (
    "<|im_start|>system\n"
    "You are a professional machine translation system.\n"
    f"Translate from {src_lang} to {tgt_lang}.\n"
    "Rules:\n"
    "1. Output ONLY the translation.\n"
    "2. Do NOT explain.\n"
    "3. Do NOT think aloud.\n"
    "4. Do NOT add notes or comments.\n"
    "5. Preserve the original meaning, tone, and style.\n"
    "6. Use natural, fluent target-language text.\n"
    "<|im_end|>\n"
    "<|im_start|>user\n"
    f"{src}\n"
    "<|im_end|>\n"
    "<|im_start|>assistant\n"
    "/no_think\n"
)

```

1.4 Notes on prompt standardization

This template was treated as part of the fixed baseline definition and was not changed across evaluation runs.

It explicitly constrains the model to:

- produce only the translation,
- avoid explanations or reasoning traces,
- preserve meaning, tone, and style,
- and generate fluent target-language output.

Because prompt wording can materially affect translation quality, this exact template should be reused in future evaluations whenever Sprint 48 results are referenced.

1.5 Evaluation setup

- Dataset: FLORES-200 devtest
- Directions: EN→CMN, CMN→EN, EN→YUE, YUE→EN
- Primary metric: COMET-22
- Secondary metrics: BLEU, chrF

2. Fixed GLM Configuration

The Sprint 48 GLM comparison used a single fixed checkpoint and quantization method.

- Model variant: GLM-4-9B
- Checkpoint: zai-org/GLM-4-9B-0414
- Quantization: 4-bit bitsandbytes

This fixed setup ensures comparability with the baseline and avoids confounding factors from changing checkpoints or inference formats.

3. Standardized FLORES-200 Results

GLM-4-9B performs slightly better than Qwen3-8B in both Mandarin directions: EN→CMN and CMN→EN. The gains are consistent across COMET, BLEU, and chrF, though mostly moderate in size. Results are more mixed for Cantonese:

- EN→YUE: Qwen3-8B is clearly stronger than GLM-4-9B across COMET, BLEU, and chrF.
- YUE→EN: the two models are close; GLM has a very small COMET advantage, but Qwen has a better chrF score.

3.1 Baseline results: Qwen3-8B

Direction	COMET	BLEU	chrF
CMN → EN	0.8797	30.7719	61.3590
EN → CMN	0.8978	50.4211	46.3901
EN → YUE	0.8783	35.6039	31.6155
YUE → EN	0.8739	29.1752	60.7115

3.2 GLM-4-9B results

Direction	COMET	BLEU	chrF
CMN → EN	0.8821	32.4995	61.7057
EN → CMN	0.9029	52.9481	46.4594
EN → YUE	0.8613	32.1900	29.5454
YUE → EN	0.8753	29.2962	59.8195

3.3 Direct Comparison Against Baseline

Direction	Qwen COMET	GLM COMET	Delta	Qwen BLEU	GLM BLEU	Delta	Qwen chrF	GLM chrF	Delta
EN → CMN	0.8978	0.9029	+0.0051	50.4211	52.9481	+2.5271	46.3901	46.4594	+0.0694
CMN → EN	0.8797	0.8821	+0.0025	30.7719	32.4995	+1.7276	61.3590	61.7057	+0.3467
EN → YUE	0.8783	0.8613	-0.0171	35.6039	32.1900	-3.4139	31.6155	29.5454	-2.0701
YUE → EN	0.8739	0.8753	+0.0014	29.1752	29.2962	+0.1210	60.7115	59.8195	-0.8920

If the project priority is balanced multilingual translation quality, GLM remains competitive. However, if English-to-Cantonese generation quality is weighted most heavily, Qwen3-8B remains the safer primary model.

This is especially important because EN→YUE is the direction most sensitive to:

- Cantonese authenticity,
- Mandarin leakage in Traditional script,
- and practical usability for Hong Kong Cantonese output.

4. Blind Human Review

To satisfy the sprint acceptance criteria, a blind human evaluation should be conducted using a representative sample of **100 EN→YUE** translations and **100 EN→CMN** translations generated by the evaluated models.

EN→YUE Evaluation Criteria

The review should assess:

- Authenticity of written Cantonese
- Appropriate use of Cantonese-specific particles, vocabulary, and grammatical structures
- Absence of Mandarin expressed in Traditional Chinese characters
- Overall fluency and translation adequacy

EN→CMN Evaluation Criteria

The review should assess:

- Translation accuracy
- Fluency
- Natural Mandarin wording and sentence structure

Automated evaluation results indicate that GLM-4-9B performs noticeably worse than Qwen3-8B on the EN→YUE translation task. Therefore, the blind human evaluation is particularly important to determine whether this performance gap is perceptible to human reviewers, whether GLM tends to generate Mandarin-like text instead of authentic Cantonese, and whether its translation quality remains sufficient for inclusion in the translation pipeline despite the lower automatic evaluation scores.

4.1 Blind Human Review Guidelines

Purpose

This review is designed to compare anonymized translation outputs from two models under a blind evaluation setup. Reviewers will not know which output comes from which model.

The review covers:

- 100 English → Cantonese (EN→YUE) samples
- 100 English → Mandarin Chinese (EN→CMN) samples

The goal is to assess:

- translation quality,
- fluency,
- adequacy,
- and, for Cantonese, authenticity versus Mandarin-in-Traditional-script leakage.

General Rules

1. Do not try to infer which model produced which output.
2. Judge only the output quality.
3. Evaluate each pair independently.
4. If both outputs are poor, still choose the better one if possible.
5. Use the notes field for any important issue, such as mistranslation, awkward phrasing, script leakage, or unnatural wording.

EN → YUE Evaluation Criteria

For English → Cantonese, the most important question is whether the output is ****genuine Cantonese**** rather than ****Mandarin written in Traditional Chinese characters****.

Evaluate the following:

1. Fluency (1–5)

How natural and readable is the output as in Cantonese?

- 5 = highly fluent and natural
- 4 = mostly fluent, minor awkwardness
- 3 = understandable but noticeably awkward
- 2 = poor fluency
- 1 = very unnatural or broken

2. Adequacy (1–5)

How well does the translation preserve the source meaning?

- 5 = complete and accurate
- 4 = mostly accurate, minor omissions
- 3 = partially accurate
- 2 = significant meaning loss/errors
- 1 = largely incorrect

3. Cantonese Authenticity (1–5)

Does the output sound like genuine Cantonese?

- 5 = clearly authentic Cantonese
- 4 = mostly Cantonese, slight formal/Mandarin influence
- 3 = mixed Cantonese and Mandarin style
- 2 = mostly Mandarin in Traditional script
- 1 = clearly not authentic Cantonese

4. Leakage Category

Choose one:

- `authentic\_cantonese`
- `mixed\_cantonese\_mandarin`
- `mostly\_mandarin\_traditional`
- `unclear`

Cantonese indicators reviewers may notice

Common Cantonese signals include:

- particles and function words such as: 嘅, 咗, 嘅, 冇, 唔, 呢, 啲, 佢, 佢哋
- Hong Kong Cantonese phrasing
- colloquial sentence structure
- natural spoken/written Cantonese rhythm

Mandarin leakage indicators

Possible Mandarin-in-Traditional-script leakage includes:

- standard written Mandarin syntax in Traditional characters
- lack of Cantonese particles/grammar where expected
- phrasing that feels formal Mandarin rather than Cantonese

EN → CMN Evaluation Criteria

For English → Mandarin Chinese, focus on overall translation quality.

Evaluate the following:

1. Fluency (1–5)

How natural and readable is Mandarin?

2. Adequacy (1–5)

How accurately does the output preserve the source meaning?

3. Naturalness (1–5)

Does the translation sound like natural Mandarin Chinese rather than overly literal or awkward text?

- 5 = very natural
- 4 = mostly natural
- 3 = acceptable but somewhat literal/awkward
- 2 = unnatural
- 1 = very poor

Optional error flags

Use notes for:

- mistranslation
- omission
- hallucination
- awkward literal wording
- grammar issues

Preference Decision

For each sample, choose:

- `A`
- `B`
- `Tie`

Select the output that is better overall for the target language and task.

Notes

Please use the notes field whenever:

- the meaning is wrong,
- the output is unnatural,
- the Cantonese feels fake or Mandarin-like,
- or both outputs are equally strong/weak for a specific reason.

4.2 Blind Human Review Protocol

Objective

This blind review compares GLM-4-9B and the baseline model on:

- 100 EN→YUE samples
- 100 EN→CMN samples

The purpose is to supplement automatic metrics with human judgment before making a final model adoption decision.

Review Design

- Each source sentence is paired with two anonymized outputs:
 - `output\_A`
 - `output\_B`
- Model identities are hidden from reviewers.
- Output order should be randomized per row.
- Reviewers should not know which output corresponds to GLM or baseline.

Sample Selection

- Draw 100 examples from the FLORES-200 devtest set for EN→YUE.
- Draw 100 examples from the FLORES-200 devtest set for EN→CMN.
- Use the same source sentence for both compared outputs.
- Preserve a hidden mapping file outside the review sheet for later analysis.

Review Fields

EN→YUE

For each sample, reviewers record:

- preferred output (`A`, `B`, or `Tie`)
- fluency score for A and B
- adequacy score for A and B

- Cantonese authenticity score for A and B
- leakage category for A and B
- optional notes

EN→CMN

For each sample, reviewers record:

- preferred output (`A`, `B`, or `Tie`)
- fluency score for A and B
- adequacy score for A and B
- naturalness score for A and B
- optional notes

Recommended Review Process

1. Randomize A/B assignment before creating the CSV.
2. Have at least one fluent Cantonese reviewer for EN→YUE.
3. If possible, use two reviewers and compare agreement.
4. Aggregate:
 - win counts,
 - average scores,
 - leakage frequency,
 - and notable qualitative issues.

Suggested Decision Logic

GLM can be recommended more strongly if:

- it wins or ties most EN→CMN samples,
- and its EN→YUE outputs are judged authentic Cantonese with low Mandarin leakage.

If GLM underperforms on EN→YUE authenticity despite good metrics, it should remain an alongside/fallback model rather than replace the baseline.

5. Pipeline Fit Assessment

5.1 QLoRA Compatibility

GLM-4-9B has already been successfully loaded and evaluated using 4-bit quantization, indicating that it is well suited for parameter-efficient fine-tuning with a QLoRA workflow. While this demonstrates strong compatibility with the planned training pipeline, a brief adapter-training smoke test is still recommended to validate the end-to-end fine-tuning process before launching large-scale training.

5.2 Tokenizer Efficiency for Cantonese

Sprint 47 research found no evidence that GLM incorporates a Cantonese-specific tokenizer or dedicated optimization for Cantonese linguistic features. This observation is consistent with the model's evaluation results, where EN→YUE translation quality lagged behind Qwen3-8B. Although GLM demonstrates adequate multilingual capability, its tokenizer and pretraining appear less optimized for Cantonese-specific vocabulary and grammatical particles, which likely contributes to the observed performance gap.

5.3 Multi-LoRA Serving Support

GLM-4-9B appears to be compatible with parameter-efficient fine-tuning (PEFT) workflows, making it a suitable candidate for adapter-based deployment. However, the serving infrastructure should be validated to ensure that it supports efficient loading, switching, and management of multiple LoRA adapters with GLM as effectively as it does for Qwen-based models. Unless significant implementation challenges emerge during integration, this should be considered a routine engineering validation rather than a limiting factor in model selection.

6. Final Recommendation

Decision

Adopt **GLM-4-9B** as an additional evaluation and fine-tuning candidate alongside **Qwen3-8B**, but retain **Qwen3-8B** as the primary baseline at this stage.

Supporting Evidence

The evaluation results indicate that GLM-4-9B outperforms Qwen3-8B on the **EN→CMN** and **CMN→EN** translation tasks while remaining competitive on **YUE→EN**. However, Qwen3-8B achieves substantially better performance on **EN→YUE**, which is the most critical direction for generating high-quality Cantonese translations. Consequently, although GLM-4-9B demonstrates sufficient overall performance to warrant further consideration, the current evidence does not support replacing Qwen3-8B as the primary baseline. Additional validation through blind human evaluation and pipeline integration testing is recommended before revisiting this decision.

Operational Recommendation

- Retain **Qwen3-8B** as the primary model for evaluation and QLoRA fine-tuning.
- Include **GLM-4-9B** as a secondary candidate for comparative evaluation, fine-tuning, and fallback deployment scenarios.
- Reassess the model selection after completing the planned blind human evaluation and a short end-to-end QLoRA integration test.